Project Report – Snake Game (Scratch)

1. Project Title

Snake Game – Arrow Key Controlled Score Game

2. Introduction

This project is a simple and interactive Snake Game created using Scratch. The player controls a snake using the arrow keys, and the aim is to avoid letting the snake's head touch its own body. Each time the snake eats a target (food), the score increases. The game ends if the snake collides with itself.

3. Objective

- To create a fun and responsive game controlled by arrow keys.

- To introduce basic game development concepts in Scratch such as movement, collision detection, and score tracking.

4. Features

- Arrow key movement (Up, Down, Left, Right)

- Score system that increases when food is collected

- Message display when player scores

- Game over condition when snake touches its own body

- High-score challenge: Users can try to beat the score of 1976

5. Working Principle

a. Snake Movement

The snake's head moves in the direction of the last pressed arrow key. Each step moves a few pixels forward creating smooth motion.

b. Body Follow Mechanism

Clones are used to represent the snake's body. Each clone follows the head with a small delay, creating the snake effect.

c. Food System

Food appears at random positions on the stage. If the snake touches the food:

- Increase score by 1

• Show a message “Scored!”

• Create a new body segment (clone)

• Move the food to a new random position

d. Game Over

If the snake’s head touches any part of the body:

• Stop all scripts

• Display “Game Over!”

6. Variables Used

• Score – stores the current score

• High Score – optional

• Direction – stores current movement direction

• Speed – speed of snake movement

7. Sprites Used

• Snake Head

• Snake Body (clone)

• Food

• Message/Score Display Area

8. Controls

• Up Arrow – Move Up

• Down Arrow – Move Down

• Left Arrow – Move Left

• Right Arrow – Move Right

9. Output

• Game runs smoothly with real-time movement

• Score increases correctly

• Displays “Scored!” whenever the snake eats food

• Shows “Game Over!” when collision occurs

10. Conclusion

This project demonstrates the basic concepts of game development using Scratch, such as sprite control, cloning, collision detection, and score handling. It provides a fun challenge for players to try beating the high score of 1976.